

Literature-implied partial answer to an LPR exponent question

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Status. This is a **literature-implied partial answer**, not a new proof packet and not a full solution of exponent-independence for the vector-valued Littlewood–Paley–Rubio de Francia property.

Source question. Hytönen–Torrea–Yakubovich, *The Littlewood–Paley–Rubio de Francia property of a Banach space for the case of equal intervals* (arXiv:0807.2981), introduce the arbitrary-interval LPR_p property and recall the known necessary conditions $LPR_p \Rightarrow \text{UMD}$ and type 2. In the introduction, lines 340–344 of the parsed source say that “it is not known whether there is an implication between LPR_p and LPR_q ” for two different exponents $p, q \in [2, \infty)$. The paper then solves the equal-length interval variant, not the arbitrary-interval exponent question.

Supporting theorem. Potapov–Sukochev–Xu, *On the vector-valued Littlewood–Paley–Rubio de Francia inequality* (arXiv:1104.2671), explicitly return to the same Banach-space LPR_p property. Their introduction cites the Hytönen–Torrea–Yakubovich necessary condition and again records that it is unknown whether LPR_p is independent of p . They then prove the upward direction. Theorem 3.1 states:

$$X \in LPR_q, \quad 2 \leq q < \infty \quad \implies \quad X \in LPR_p \quad \text{for every } q \leq p < \infty.$$

Identification. This theorem gives a positive implication between two different LPR exponents in the direction from a smaller exponent to every larger exponent. It is therefore a literature answer to the upward half of the source’s two-exponent question.

Scope. The supporting theorem does not prove the reverse implication, so full exponent-independence remains open here. It also does not settle the broader converse problem $\text{UMD} + \text{type } 2 \Rightarrow LPR_p$ for arbitrary intervals. The run contains a serious no-promotion attempt, `attempts/1104.2671_lpr_umd_type2_bootstrap_barrier.md`, explaining why the equal-length theorem does not formally bootstrap to arbitrary intervals: the missing step is a dimension-free R -bounded summation over dyadic scales.

Search evidence. The cheap run indexes were searched for arXiv:0807.2981, arXiv:1104.2671, Littlewood–Paley–Rubio, Rubio de Francia, LPR_p , LPR_q , UMD, and type 2. The relevant existing run memory is the no-promotion attempt on arXiv:1104.2671, not a durable packet for this partial literature answer. External arXiv/web search located arXiv:1104.2671 as the decisive later source and did not locate a full arbitrary-interval exponent-independence solution.

References

- [1] T. P. Hytönen, J. L. Torrea, and D. V. Yakubovich, *The Littlewood–Paley–Rubio de Francia property of a Banach space for the case of equal intervals*, arXiv:0807.2981.
- [2] D. Potapov, F. Sukochev, and Q. Xu, *On the vector-valued Littlewood–Paley–Rubio de Francia inequality*, arXiv:1104.2671.